



200 series

SH-pH200

pH digital sensor

User manual

Reading instructions

Information for users

Thank you very much for choosing to use our SH-pH200 pH online sensor (hereinafter referred to as: pH200 sensor). Before using this product, please read this user manual carefully. This manual covers important information and data for the use of the product, which must be strictly observed by the user to ensure the normal operation of the pH200 sensor. The information will help you to use the product correctly and to obtain accurate analysis results.

General description

This manual provides a detailed description of the installation, commissioning, operation and maintenance of the pH200 sensor, as well as describing the measurement principle, instrument composition and performance characteristics of the pH200 sensor.

This manual instructs the user on the proper installation and operation of the pH200 sensor, as well as preventive maintenance work on the pH200 sensor to guarantee the continuous and reliable operation of this sensor.

The products described in this manual have been rigorously inspected before leaving the factory to ensure top quality. In order to ensure safe, high-quality operation and to obtain correct analysis results, the user must strictly follow the instructions described in this manual.

This manual provides all the information required for the proper use of the pH200 sensor. It provides an accurate reference for use by technicians with specialized training or knowledge of the instrument's operational controls. Before operating the instrument, make sure that the safety and warning messages mentioned in this manual are correctly understood and applied to the actual operation.

Delivery and transport

Shipment requirements are in accordance with the terms of the ordering contract.

Please read the appropriate information on the packaging material carefully when opening the box to ensure that the opened goods are complete and undamaged. Please try to retain the outer packaging of the product in case you need to return the meter or parts.

Warranty and maintenance

1) SYSTEALOG warrants all Products manufactured to be free from defects in material and workmanship under normal use and service. SYSTEALOG's obligation under any warranty is limited to and Buyer's exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEALOG's factory, of any defective parts, when returned to SYSTEALOG by Buyer, transportation prepaid, which SYSTEALOG's examination discloses to have been factory defective. The time limit of this warranty is 12 (twelve) months from date of shipment of new Products unless otherwise specified by SYSTEALOG. SYSTEALOG shall not be liable for consequential damages, in any event. SYSTEALOG will decide at its only discretion whether to repair or replace a defective part.

For wearing parts and/or in contact with aggressive substances SYSTEALOG reserves to reduce the warranty period accordingly to the specific application.

2) Repaired Products, all repair work done by SYSTEALOG is warranted to be free from defects in material or workmanship under normal use and service. SYSTEALOG's obligation under this warranty is limited to, and the exclusive remedy hereunder shall be, repair or replacement, Ex Works SYSTEALOG's factory, of any defective parts installed by SYSTEALOG, when returned to SYSTEALOG, transportation prepaid, which SYSTEALOG's examination discloses to have been factory defective. The time limit of this warranty is three (3) months from date of shipment. SYSTEALOG shall not be liable for consequential damages, in any event. SYSTEALOG will decide at its only discretion whether to repair or replace a defective part.

3) THESE WARRANTIES ARE IN LIEU OF ANY WARRANTY OF MERCHANTABILITY, OR FITNESS, OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

4) Expendable Items: this Warranty does not cover expendable or consumable items such as lamps, fuses, parts in contact with fluids, etc.

5) SYSTEALOG has reasonable period of time for replacements/repairs, and cannot liable for any compensation may arise in relation to the Product

6) SYSTEALOG only obligation in case of defects, lack of quality or non-conformity of the Products will be repairing or replacing the defective parts. It is agreed that the above-mentioned warranty (i.e.: the obligation to repair or replace the defective parts) is in lieu of any other legal guarantee or liability with the exclusion of any other SYSTEALOG liability (whether contractual or non-contractual) which may anyhow arise out of or in relation with the Products supplied (e.g. compensation of damages, loss of profit, recall campaigns, etc.).

8) The warranty on goods supplied by SYSTEALOG is void after 12 months for new Products, if not otherwise specified, 3 months for repairs from delivery date, and in case of one of the following conditions:

- a) Due to improper use (water ingress, corrosion, fire, strong electricity strung in, etc.);
- b) Damage caused by force majeure (earthquakes, lightning strikes, floods, etc.);
- c) Unauthorized alterations within the product without permission;
- d) Failure to use the product in accordance with the user's manual and training provisions, causing damage to the product.
- e) On products not manufactured by SYSTEALOG but sold by SYSTEALOG as results of a distribution contract or as occasional resale, apply the warranty conditions of the manufacturer of the product.
- f) Any warranty request, within specified limits, must be notified, the SYSTEALOG using the form "Warranty Claim" by fax, mail or registered mail, within seven (7) days from the event of the defect. Failing such notification, the Purchaser's right to claim the defects will be forfeited.

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1 Instrument overview

1.1 Instrument Introduction

The pH-200 sensor is capable of monitoring both temperature and pH parameters in water in real time. The sensor consists of electrodes (including temperature and pH electrodes), circuits (internal) and structural parts, and its appearance is shown in Figure 1.1.

This product adopts the glass electrode method to measure the pH of the water body, including the detection of temperature and pH indicators. Temperature measurement adopts NTC temperature sensor, the voltage detected at different temperatures is processed by the circuit and converted to the corresponding temperature value by the instrument. pH measurement adopts glass electrode, the electrode potential is collected, and the corresponding potential is converted to pH value by the Nernst equation. At the same time, the pH value is temperature compensated according to the measured temperature value to obtain a more accurate pH value.

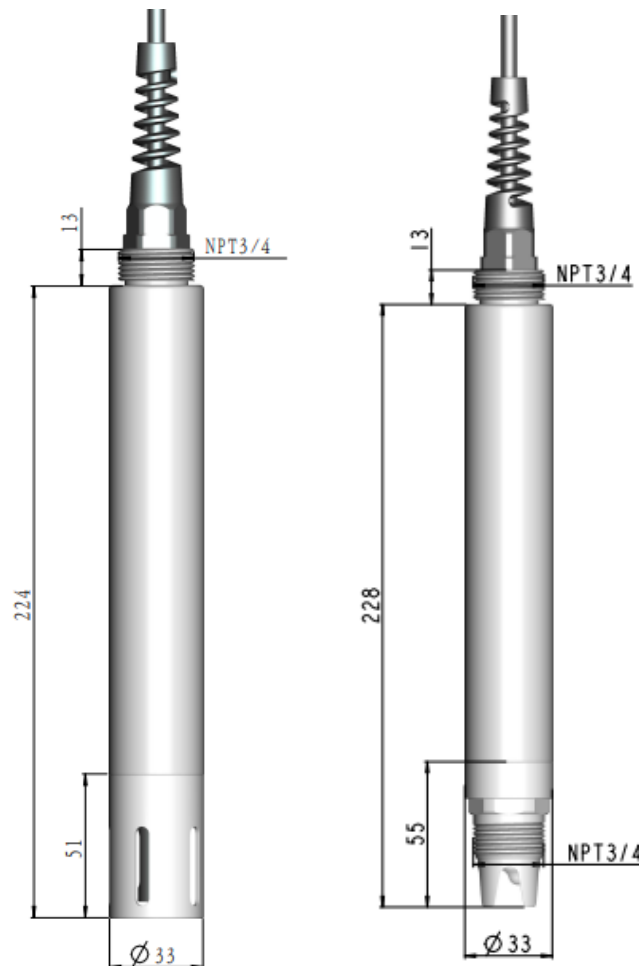


Figure 1.1 Schematic of the appearance of the pH online sensor
(optional lower protective cap form for installation and use)

1.2 Product features

- Industrial grade composite pH electrode for harsh working environments.
- NTC temperature measurement, measurement accuracy up to $\pm 0.1^{\circ}\text{C}$.
- With automatic temperature compensation function, applicable to a wide range of environments.
- Corrosion-resistant housing, waterproof grade IP68, can work underwater for a long time.
- RS485 communication interface with standard Modbus protocol for easy integration.
- Intelligent data sensor with calibration record keeping and abnormal alarm function.

1.3 Main uses and scope of application

Widely used in surface water, industrial wastewater and municipal sewage online monitoring.

1.4 Environmental conditions of use

Temperature: (0-60) $^{\circ}\text{C}$; Pressure: (0-4) bar

2 Technical specification

Working principle	Glass electrode method
Measurement range	pH (0~14); Temperature (0~50) °C
Repeatability	<0.1 pH
Drift (24h)	<0.1 pH
Resolution	0.01 pH
Response time	<10 s
Protection class	IP68
Comm. interface	RS485, standard Modbus protocol, baud rate 9600
Dimensions	($\phi 33 \times 224$) mm / ($\phi 16 \times 180$) mm, cable 10 m (customizable)
Materials	316 Stainless-steel/PVC/POM/Titanium alloy available
Weight	0.5 kg
Mounting threads	NPT3/4 thread on top, NPT3/4 male thread on bottom available
Operating voltage	9~36 Vdc
Power absorption	0.2 W
Calibration period	1-3 months
Max. depth	30 meters
Operating temp.	(0~50) °C

Table 2.1

3 Use and operation

3.1 installation instructions

The installation of the pH sensor has a large impact on its measurement, and the environment in which it is installed and used needs to meet the following requirements:

- When installing the sensor, make sure that the sensor area is not damaged by external collisions, cuts and pressure.
- The minimum water level should be submerged after installation to prevent the electrode from contacting the liquid to be measured when the water level is too low.
- For submerged installation, the sensor must be mounted on the mounting bracket, and it is not acceptable to suspend the sensor in water with the pH sensor cable.

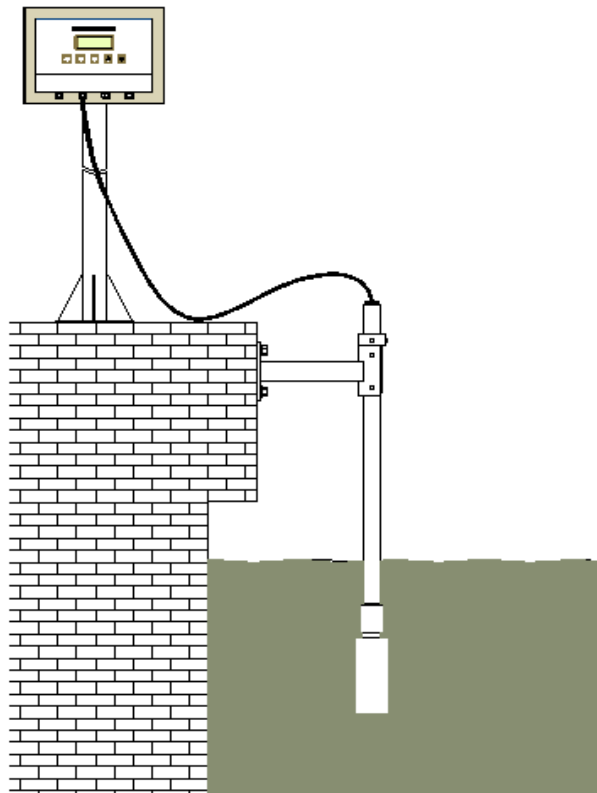


Figure 3.1 Schematic diagram of submerged installation

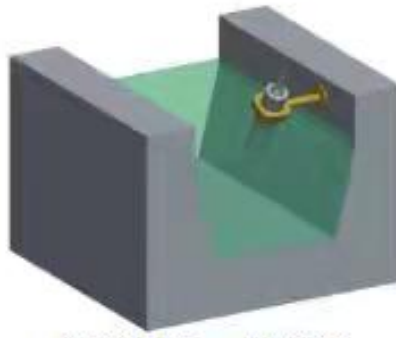


Figure 3.2 Schematic of sensor in channel installation

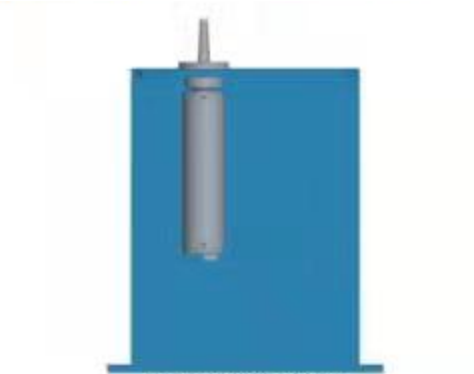


Figure 3.3 Schematic of sensor in pool installation



Figure 3.4 Schematic of 45° pipe installation

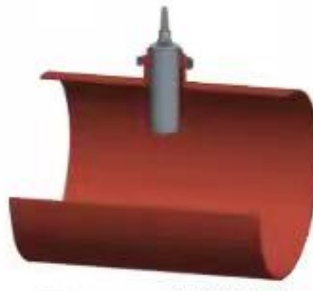


Figure 3.5 Schematic of 90° pipe installation

3.2 Wiring instructions

Sensor type	Wire color	Meaning	Connection type
pH200 online sensor	red	Power positive (+)	Power supply
	black	Power negative (-)	Power supply
	blue	485A	RS-485 communication
	yellow-brown	485B	RS-485 communication
	violet	PE	grounding

Table 3.1 Wiring description

Sensor wiring, strip the end of the cable to reveal 5 wires, please check the table 3.1 were connected to the corresponding port, any short circuit may cause the instrument burned, data loss or program damage! Confirmation of the correct wiring can be connected to the power to use.

WARNING: The PE probe wire must be connected to the PE terminal of the controller.

CAUTION: Do not connect the PE probe wire to the main power terminals (L and N), otherwise serious damage to the probe or controller may occur. Carefully check the probe wire connection before turning on the controller.

3.3 Communication protocol

baud	9600
data bit	8
stop bit	1
parity check	None
slave address	0x32

Table 3.2 Serial Port Default Parameter Configuration

Modbus-RTU mode, data transfer size end description:

data type	Big and small end model
char (computing)	the main points
uint16	the main points
uint32	Word-ordered little end, byte-ordered big end within word (CDAB)
Float (encoding format IEEE754)	Word-ordered little end, byte-ordered big end within word (CDAB)

Fault code	Reason for failure
1ul<<0	Abnormal pH signal
1ul<<1	Excessive temperature difference between calibration points
1ul<<2	Calibration slope anomaly
1ul<<3	Excessive temperature difference between calibration points
1ul<<4	Calibration slope anomaly
1ul<<8	Temperature measurement abnormality

Table 3.3 Fault table

Note:

<< is a left-shift operator that first converts the alarm code to binary, with each bit representing an alarm state

For example, 44 is converted to 00000000 00000000 00000000 00101100, which corresponds to fault codes 1ul<<2,1ul<<3,1ul<<5

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Device SN number	ASCII	--	--	12	0x7530	
software version number	ASCII	--		32	0x7536	
Probe Type	uint16		--	2	0x7546	Fixed at 4 for pH
Probe Status	uint16	--	0-5	2	0x7566	0: Measurement Three-point calibration 1: pH1 is being calibrated 2: pH2 is being calibrated 3: pH3 is being calibrated Two-point calibration 4: pH1 is being calibrated 5: pH2 is being calibrated
error code	uint32	--	--	4	0x7567	Normal state is 0, see fault table for details
pH value	float	--	--	4	0x7569	
temp	float	°C	0-50	4	0x756b	
pH voltage	float			4	0x757f	
Temperature voltage	float			4	0x7573	
pH calibration voltage value 1 (three-point calibration)	float	mV		4	0x7575	
Temperature at pH calibration 1 (three-point calibration)	float	°C	0-50	4	0x7577	
pH calibration voltage value 2 (three-point calibration)	float	mV		4	0x7579	

Functionality	Typology	Meas. unit	Range of values	Byte count	Input register address	Note
Temperature at pH calibration 2 (three-point calibration)	float	°C	0-50	4	0x757b	
pH calibration voltage value 3 (three-point calibration)	float	mV		4	0x757d	
Temperature at pH calibration 3 (three-point calibration)	float	°C	0-50	4	0x757f	
pH calibration voltage value 1 (two-point calibration)	float	mV		4	0x7581	
Temperature at pH calibration 1 (two-point calibration)	float	°C	0-50	4	0x7583	
pH calibration voltage value 2 (two-point calibration)	float	mV		4	0x7585	
Temperature at pH calibration 2 (two-point calibration)	float	°C	0-50	4	0x7587	

Table 3.4 Input register protocols

R/W properties: phrase marked by pause

Functionality	Read and Write Properties	Typology	Meas. unit	Range of values	Occupied byte	Holding Register Address	Note
Device address	Read/Write	uint16	--	0 to 247	2	0x9c40	The default value is 0x32, the device is reset after writing, so it must be written to a single register.
Baud	Read/Write	uint32	--	2400. 4800. 9600. 14400. 19200. 38400. 57600. 115200.	4	0x9c41	Default value 9600
Restore Factory Settings	write	uint16	--	0-1	2	0x9c51	Automatic reset after writing
Restart SW	write	uint16	--	0-1	2	0x9c53	Automatic reset after writing
Number of calibration points	write	uint16	--	2-3	2	0x9c56	
pH calibration time1 (three-point calibration)	Read/Write	uint32	s	--	4	0x9c62	UNIX timestamp (calibration starts after writing)
pH calibration value1 (three-point calibration)	Read/Write	float	--	0-14	4	0x9c64	
pH calibration time2 (three-point calibration)	Read/Write	uint32	s	--	4	0x9c66	UNIX timestamp (calibration starts after writing)
pH calibration value2 (three-point calibration)	Read/Write	float	--	0-14	4	0x9c68	

pH calibration time 3 (three-point calibration)	Read/Write	uint32	s	--	4	0x9c6a	UNIX timestamp (calibration starts after writing)
pH calibration value 3 (three-point calibration)	Read/Write	float	--	0-14	4	0x9c6c	
pH calibration time1 (two-point calibration)	Read/Write	uint32	s	--	4	0x9c72	UNIX timestamp (calibration starts after writing)
pH calibration value1 (two-point calibration)	Read/Write	float	--	0-14	4	0x9c74	
pH calibration time2 (two-point calibration)	Read/Write	uint32	s	--	4	0x9c76	UNIX timestamp (calibration starts after writing)
pH calibration value2 (two-point calibration)	Read/Write	float	--	0-14	4	0x9c78	
Temperature calibration value	Read/Write	float	°C	--	4	0x9c7a	

Table.3.5 Holding registers

3.4 MODBUS-RTU protocol example (address default 0x32)

3.4.1 Read the measured value

Send: 32 04 75 69 00 04 3e 1a

0x32:Device address

0x04:Read input register function code

0x7569:Input register start address

0x04:read 4 registers

0x3e1a:check digit

Received: 32 04 08 33 33 40 f3 4c 10 42 08 b8 ad

0x32:Device address

0x04:Read input register function code

0x08:8 bytes

0x333340f3: pH 0x40f33333 (7.6)

0x4c104208:temperature 0x42084c10 (34.07)

0xb8ad:Check digit

3.4.2 pH1 calibration

Send: 32 10 9c 62 00 04 08 65 35 64 f9 51 ec 40 80 ba f8

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c62:Write register start address

0x04:Write 4 registers 0x08:Write 8 bytes

0x653564f9: timestamp 0x64f96535 (1694065973) 2023-09-07 13:52:53

0x51ec4080: calibration value 0x408051ec (4.01)

0x07bd:Check Digit

Received: 32 10 9c 62 00 04 4b 87

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c62:Write register start address

0x04:Write to 4 registers

0x4b87:Check Digit

3.4.3 pH2 calibration

Send: 32 10 9c 66 00 04 08 69 50 64 f9 85 1f 40 db f6 5c

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c66:Write register start address

0x04:Write to 4 registers

0x08:Write 8 bytes

0x695064f9: timestamp 0x64f96950 (1694067024) 2023-09-07 14:10:24

0x851f40db: calibration value 0x40db851f (6.86)

0xf65c:Check digits

Received: 32 10 9c 66 00 04 0a 46

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c66:Write register start address

0x04:Write to 4 registers

0x0a46:Check digit

3.4.4 pH3 calibration

Send: 32 10 9c 6a 00 04 08 6a b4 64 f9 e1 48 41 12 6d 71

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c6a:Write register start address

0x04:Write to 4 registers

0x08:Write 8 bytes

0x6ab464f9:timestamp 0x64f96ab4 (1694067380) 2023-09-07 14:16:20

0xe1484112: calibration value 0x4112e148 (9.18)

0x6d71:check digit

Received: 32 10 9c 6a 00 04 ca 45

0x32:Device address 0x10:Write multiple registers function code 0x9c6a:Write register start address

0x04:Write 4 registers 0xca45:Checksum code

3.4.5 Temperature calibration

Send: 32 10 9c 7a 00 02 04 00 00 41 a0 4d 9e

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c7a:Write register start address

0x02:Write to 2 registers

0x04:write 4 bytes

0x000041a0:Thermometer measurement 0x41a00000(20)

0x4d9e:check digit

Received: 32 10 9c 7a 00 02 4b 82

0x32:Device address

0x10:Write Multiple Register Function Code

0x9c7a:Write register start address

0x02:Write to 2 registers

0x4b82:check digit

3.4.6 *Modify calibration points*

Send: 32 06 9c 56 00 03 02 48

0x32:Device address

0x06:Write Single Register Function Code

0x9c56:Write register address

0x0003:Three-point calibration

0x0248:Check digit

Received: 32 06 9c 56 00 03 02 48

0x32:Device address

0x06:Write Single Register Function Code

0x9c56:Write register address

0x0003:Three-point calibration

0x0248:Check digit

4 Calibration and maintenance

4.1 Sensor calibration

The pH sensor as default is set to three-point calibration; it has to be connected to the controller or host computer and the following steps are followed to complete the three-point calibration.

4.1.1 First calibration point (pH=4)

Nominal 4 pH standard solution point calibration: put the sensor into the nominal 4 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 1" at the current standard pH concentration (against the current temperature to get the standard value), and then in the "pH: calibration time 1" to modify the corresponding time stamp for calibration, or in the controller at the "pH: calibration point 1" enter the current standard pH concentration (against the current temperature to get the standard value). calibration time 1" to modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 1" at the current standard liquid pH concentration (against the current temperature to get the standard value), and then click "calibration". Wait for about 10s to complete the calibration of the nominal 4 pH standard solution point.

4.1.2 Second calibration point (pH=7)

Nominal 7 pH standard solution point calibration: put the sensor into the nominal 7 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 2" at the current standard pH concentration (against the current temperature to get the standard value), and then in the "pH calibration time 2" at the corresponding time stamp to modify the calibration, or in the controller "pH: calibration point 2" at the current standard liquid pH concentration (against the current temperature to get the standard value), and then click "calibration". Wait for about 10 seconds to complete the calibration of the nominal 7 pH standard solution point.

4.1.3 Third calibration point (pH=9 or 10)

Nominal 9 or 10 pH standard solution point calibration: put the sensor into the nominal 9 or 10 pH buffer solution, wait until the test value is completely stable (generally 5-10min), in the "pH calibration value 3" at the current standard pH concentration (against the current temperature to get the standard value), and then in "pH: calibration time 3" to modify the corresponding time stamp for calibration, or in the controller at the "pH: calibration point 3" enter the current standard pH concentration (against the current temperature to get the standard value). calibration time 3" to modify the corresponding time stamp for calibration, or in the controller "pH: calibration point 3" at the current standard liquid pH concentration (against the current temperature to get the standard value), and then click "calibration". Wait for about 10s to complete the calibration of the nominal 9 or 10 pH standard solution point.

4.1.4 Two points calibration procedure

If the site needs to use two-point calibration mode, first switch to the number of calibration points for 2, and then refer to the three-point calibration mode to arbitrarily select two points for calibration, and note that the lower pH calibration point of the calibration liquid corresponds to the calibration value of pH (2-point calibration) 1. If the site environment is acidic, it is recommended to select the 4 pH standard

solution (calibration point 1) and the 7 pH standard liquid (calibration point 2); if the site environment is alkaline, it is recommended to select the 7 pH calibration liquid (calibration point 1) and the 9 or 10 pH calibration liquid (calibration point 2).

4.2 Sensor maintenance

Sensor maintenance consists mainly of cleaning and calibration.

4.2.1 Cleaning

In order to prevent water impurities and microorganisms adhering to the electrode surface caused by inaccurate measurements, it is recommended to clean once a week. When cleaning, remove the sensor from the water, use a clean non-woven cloth and distilled water to remove the dirt on the surface of the temperature sensor and pH electrode, and be careful not to damage the electrode with too much force.

4.2.2 Calibration

The calibration frequency is mainly depending on the water matrix and specific site conditions; for the best accuracy, it is usually recommended to perform a calibration once a week after cleaning.

4.3 Sensor storage

The pH sensor is shipped covered by a tap including a KCl 3M buffer solution; in order to use it on site, please remove the tap.

When the instrument you purchased is not to be installed urgently or needs to be put aside for a longer period of time, please store it properly and do not put it in the following places:

- Ambient temperature below 5°C or above 40°C;
- Direct sunlight;
- Next to the heater;
- The environment is vibrating and dusty.

Keep the pH electrode in its moist protective sleeve as long as possible before installation. When the instrument you have purchased has been installed but will not be used for a longer period of time, please clean the electrode with DI water, keep the pH probe protective sleeve in place, remove the probe sleeve when activated and close the pH sensor with the same shipment tap filled up by KCl 3 M buffer solution.

4.3.1 Preparation of pH buffer storage solution

Method of Preparing Saturated Potassium Chloride Solution:

1. Weight Potassium Chloride Solid: Use a balance to weight 34.2 grams of potassium chloride solid.
2. Dissolve Potassium Chloride: Pour the weighed potassium chloride solid into a 300 ml measuring cup and add 100 ml of distilled water.
3. Stirring: Use a glass rod to thoroughly stir until the potassium chloride is completely dissolved, resulting in a saturated potassium chloride solution.

Notes:

- Temperature Control: Potassium chloride has a solubility of 34.2 g at 20 degrees Celsius, so preparation should be done at 20 °C.



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- Stirring: Ensure thorough stirring to guarantee complete dissolution of potassium chloride and avoid local oversaturation.
 - Safety: Wear appropriate protective gear during preparation to avoid skin contact and inhalation of potassium chloride dust.